Supplementary Information for: Resolution and Cost in Sequential Evaluation of Reinforcement Learning Algorithms: A Design Audit

Status. This supplement accompanies the v3 primary manuscript. It contains diagnostic, scope-expansion, and resolution-cost analyses that were not pooled into the preregistered v3 15-comparison family. It does not change the v3 estimand, alpha allocation, stopping rule, or primary decisions. Every numeric statement below is traceable to a stored report, JSON summary, CSV file, or audit script in the submission archive.

S1. Evidence hierarchy and labeling

The package uses four evidence labels. Formal refers to the conditional finite-look proposition and its algebraic corollaries. Primary v3 refers to the preregistered five-task, three-algorithm family with 20 sealed test seeds. Diagnostic refers to analyses run to understand calibration, sensitivity, or resolution cost without changing the primary estimand. Repeatability refers to the same-machine cross-script replay; it is not independent reproducibility. The supplement never upgrades a diagnostic into a primary claim or a same-machine replay into an external replication.

Table S1. Evidence hierarchy and labeling.

Evidence item	Status in the paper	Primary source
Conditional finite-look coverage proposition	Formal methodological result under registered assumptions	Preregistration and proof materials (Online Resource 5)
Deterministic theorem-number audit	Algebraic and boundary-value check; not a probabilistic proof	Theory audit and tests (Online Resource 5)
Independent theory reimplementation	Same-workspace formula cross-check; not external statistical review	Independent theory audit (Online Resource 5)
Independent raw-data recomputation	Same-workspace recomputation of all 15 primary decisions	Recomputed results and audit code (Online Resource 2 and Online Resource 5)
15 v3 comparisons at 20 sealed seeds	Primary v3 evidence	Primary data and preregistration (Online Resource 2)
Calibration grid	Monte Carlo diagnostic	Calibration report and data (Online Resource 3)
Normalization, margin, horizon, episode, and tuning checks	Post-hoc or prospective supplementary diagnostics	Sensitivity reports and data (Online Resource 3)
Partial-observation and continuous-control runs	Separately scoped execution diagnostics	Scope reports and data (Online Resource 3)
Queue follow-up, v4, and v5	Same-family resolution-cost diagnostics	Extension reports and data (Online Resource 3)
v6 task-family extension	Two additional fixed discrete tasks; same-machine scope diagnostic	Preregistration, report, data, and audit (Online Resource 3)
Cross-script replay	Same-machine repeatability	Replay summary and code (Online Resource 2 and Online Resource 5)
Seed-dependence stress test	Controlled assumption-failure diagnostic	Dependence diagnostic report and data (Online Resource 4)
Known-block fallback	Controlled mitigation diagnostic under known independent blocks	Fallback diagnostic report and data (Online Resource 4)

S2. Calibration grid

The calibration grid evaluates the registered unpaired bounded interval under eight bounded distribution families, five true mean differences from -0.20 to 0.20, two alpha settings, four looks at 5, 10, 15, and 20 observations, and 3,000 Monte Carlo replicates per cell. The grid therefore contains 80 cells and 240,000 replicates. The two alpha settings are single-comparison alpha 0.05 and the v3 comparison allocation 0.05/15.

Table S2. Calibration-grid summary by alpha allocation.

Alpha setting	Cells	Minimum simultaneous coverage	Maximum observed false-superiority rate	Mean seed cost	Cells with a terminal decision
0.05	40	1.0000	0.0000	20.00	0
0.05/15	40	1.0000	0.0000	20.00	0

These are Monte Carlo summaries for the disclosed grid. They do not establish coverage for arbitrary bounded distributions, RL training-seed distributions, or a different stopping schedule. The raw cell-level output and Wilson intervals, generation details, and distribution definitions are provided in Online Resource 3.

S2.1 Seed-dependence assumption stress test

The coverage theorem assumes independence across sealed seed metrics within each algorithm. To make this limitation concrete, we generated two independent arms whose values are Bernoulli(0.5) but repeated within blocks of size 1, 2, 5, 10, 20, or 50. The true difference is zero, and the same four looks, alpha 0.05, 0.10 practical margin, and maximum of 200 observations are used in every condition. Each condition has 5,000 replicates. Block size 1 is the independent calibration reference; larger blocks reduce the effective number of independent observations without changing the nominal sample count.

Table S3. Seed-dependence stress-test results.

Block size	Effective blocks at n=200	Simultaneous coverage	95% Wilson interval	False-superiority rate	95% Wilson interval	Mean terminal seeds
1	200	1.0000	[0.9992, 1.0000]	0.0000	[0.0000, 0.0008]	200.00
2	100	0.9980	[0.9963, 0.9989]	0.0000	[0.0000, 0.0008]	200.00
5	40	0.9304	[0.9230, 0.9371]	0.0162	[0.0131, 0.0201]	197.74
10	20	0.7552	[0.7431, 0.7669]	0.1414	[0.1320, 0.1513]	180.91
20	10	0.4752	[0.4614, 0.4891]	0.3328	[0.3199, 0.3460]	159.16
50	4	0.1244	[0.1155, 0.1338]	0.7756	[0.7638, 0.7869]	101.67

This is an assumption stress test, not an RL experiment or a correction for dependent training. It shows why exact same-machine replay cannot establish the independence premise and why the conditional coverage language must remain in the manuscript. The generator and machine-readable output are provided in Online Resource 4.

S2.2 Known-block dependence fallback

When the block partition is known and block means are independent, Proposition 3 gives a conservative fallback: discard partial trailing blocks and apply the same interval to complete block means. The table uses the same 5,000 replicates as the preceding stress diagnostic and compares the raw-count interval with the block-aware interval.

Table S4. Known-block fallback diagnostic.

Block size	Effective blocks by look	Naive coverage	Fallback coverage	Naive false superiority	Fallback false superiority	Naive mean terminal n	Fallback mean terminal n
1	[50, 100, 150, 200]	1.0000	1.0000	0.0000	0.0000	200.00	200.00
2	[25, 50, 75, 100]	0.9982	1.0000	0.0002	0.0000	199.97	200.00
5	[10, 20, 30, 40]	0.9376	1.0000	0.0124	0.0000	198.25	200.00
10	[5, 10, 15, 20]	0.7706	1.0000	0.1370	0.0000	181.58	200.00
20	[2, 5, 7, 10]	0.4646	1.0000	0.3338	0.0000	158.16	200.00
50	[1, 2, 3, 4]	0.1264	1.0000	0.7812	0.0000	99.66	200.00

The fallback's estimated coverage was 1.0000 and its observed false-superiority rate was 0.0000 in every tested block condition. This is a Monte Carlo observation under the known independent-block model, not a proof for unknown RL dependence. The effective sample-size reduction explains why the fallback made no early decisions in this scenario. Its report, raw output, and deterministic tests are provided in Online Resource 4.

S3. Sensitivity and selection diagnostics

The following checks reuse stored rows or use separately disclosed fixed configurations. They were not used to redefine the v3 hypothesis or to select the reported primary result.

Table S5. Sensitivity and selection diagnostic map.

Diagnostic	Changed quantity	Stored observation	Interpretation limit
Normalization sensitivity	One tighter and one wider bound per task, 11 schemes total	All 11 schemes retain 15/15 maximum-budget inconclusive decisions	Robustness for the stored rows; not a new estimand or universal normalization rule
Practical-margin sensitivity	Margin values 0.00, 0.025, 0.05, 0.10, 0.20, 0.30, 0.50	Stored v3 rows remain 15/15 maximum-budget inconclusive at the tested margins	Descriptive sensitivity; not a universal practical margin or power curve
Training horizon	Fixed v3 selections, 16,384 environment steps	All 15 comparisons remain maximum-budget inconclusive	Horizon is part of the estimand; this does not prove general horizon sufficiency
Evaluation aggregation	Five versus ten episodes per checkpoint	All 15 decisions remain maximum-budget inconclusive	Does not establish a universal episode count

Diagnostic	Changed quantity	Stored observation	Interpretation limit
Tuning strength	Five versus ten versus twenty validation seeds with disjoint diagnostic tests	Four of 15 selections change at ten seeds; seven of 15 change at twenty seeds relative to v3	Selection is configuration-conditional; these are observed-sample diagnostics, not probabilities of global optimality
Validation row bootstrap	Joint resampling of the five observed validation rows	Frequencies quantify observed-sample selection instability	Not a population selection probability or a confirmatory result
Validation-selected alpha spending	Four valid schedules selected by a validation bootstrap	No observed coverage failure or false-superiority decision in four bounded scenarios; selected cost was not uniformly best	Conditional selection diagnostic; not a universal efficiency claim or a replacement for v3

The prospective-within-workspace twenty-seed extension provides a second, stronger tuning-budget check. It evaluated the same four candidates for each of the 15 cells on validation seeds 11000--11019, disjoint from v3 and the earlier diagnostics. Seven of 15 selected configurations changed relative to the v3 five-seed selection: CartPole/PPO, CartPole/DQN, Acrobot/PPO, Acrobot/A2C, NonstationaryQueue/PPO, NonstationaryQueue/DQN, and NonstationaryQueue/A2C. This 7/15 rate is an observed fixed-family selection-sensitivity result. It is not a population probability, an optimality claim, or evidence for a universal twenty-seed rule. The frozen design, raw rows, audit, and report are provided in Online Resource 3.

S3.1 Alternative intervals and pairing dependence

An additional diagnostic re-used one stored Acrobot pilot sequence to compare paired and unpaired interval constructions under the same finite-look schedule. It is intentionally outside the v3 confirmatory family. The paired normal and paired bootstrap rows re-assigned DQN outcomes to 5,000 deterministic permutations; the unpaired methods use the two marginal samples and are invariant to that re-assignment. The terminal decision counts were:

Table S6. Alternative interval and pairing decisions.

Interval construction	B superior	Precision only	Maximum-budget inconclusive	Mean seeds used
Paired normal	16	121	4,863	9.867
Paired bootstrap	143	222	4,635	9.642
Paired empirical Bernstein	0	0	5,000	10.000
Paired Hoeffding	0	0	5,000	10.000

The changing paired counts are a sensitivity-to-pairing observation, not a false-positive estimate: the same stored outcomes are being re-paired. A separate unpaired empirical-Bernstein simulation found simultaneous coverage of 1.0000 in every tested cell and no earlier resolution at 5--20 seeds, with the lower 95% Monte Carlo bound reported as 0.99923. Those results are exploratory and do not justify replacing the preregistered unpaired Hoeffding interval without a new finite-sample/sequential proof and preregistration.

The source artifacts and diagnostic report are provided in Online Resource 3. This subsection does not support claims that paired procedures are invalid in general or that the candidate empirical-Bernstein method is more efficient.

The reports retain the exact alternative bounds, seed ranges, configuration IDs, and recomputed decision records. The primary manuscript therefore reports these analyses as sensitivity evidence rather than as additional confirmatory comparisons.

S3.2 Validation-selected alpha spending

Corollary 5 applies the same conditioning logic to a finite set of valid alpha-spending schedules. The diagnostic defines four schedules with the same comparison-level alpha: power-series, front-loaded, late-loaded, and finite- look minimax. A bootstrap estimate of stopping cost from a validation sequence selects one schedule, and a fresh sealed sequence is then evaluated under that schedule. The looks are 50, 100, 150, and 200 observations, with practical margin 0.05 and target width 0.20. The validation sequence contains 200 observations per algorithm so every candidate look is represented without extrapolating the validation sample.

The diagnostic uses four bounded scenarios and 2,000 Monte Carlo repetitions per scenario. There are no observed simultaneous-coverage failures or false-superiority decisions. The selected mean costs and the best fixed- schedule costs are:

Table S7. Validation-selected alpha-spending diagnostic.

Scenario	True difference	Selected mean cost	Best fixed mean cost	Selected coverage	False-superiority rate
Null Bernoulli	0.000	200.00	200.00	1.0000	0.0000
Large effect	0.900	50.00	50.00	1.0000	0.0000
Skewed effect	0.400	121.53	119.88	1.0000	0.0000
Two-point mixture	-0.320	175.20	173.98	1.0000	0.0000

The result supports the conditional selection argument for the listed bounded scenarios, while showing that validation-selected spending is not automatically more efficient than the best fixed schedule. It is a post-v3 statistical diagnostic, not an RL experiment, a new confirmatory family, or evidence of a universal cost-saving rule. The machine-readable output and generator are provided in Online Resource 3.

S4. Scope-expansion diagnostics

S4.1 Fixed-mask partial observation

The separate fixed-mask run uses MaskedCartPole-v1 and MaskedAcrobot-v1, with the disclosed velocity coordinates replaced by zero. It contains 240 rows: 120 validation rows and 120 selected-test rows, with six comparisons. All six comparisons are maximum-budget inconclusive at 20 test seeds. The wrapper tests, preregistration, raw rows, and replay output are provided in Online Resource 3. This is a fixed-mask execution pressure test, not evidence for general POMDP performance.

S4.2 Continuous-action tasks

The Pendulum supplement contains 80 rows and one PPO--A2C comparison. Its estimate is -0.0009 with interval [-0.8757, 0.8739], and the comparison is maximum-budget inconclusive. The separately preregistered MountainCarContinuous-v0 scope expansion also contains 80 rows and one PPO--A2C comparison; the normalized means are 0.499982 and 0.499859, with estimate 0.0001233 and interval [-0.8747, 0.8749]. Both are execution-scope diagnostics, not continuous-control rankings or external replications.

S4.3 Controlled queue validity

The custom NonstationaryQueue-v0 task has capacity 10, horizon 200, a rate switch at step 100, arrival probabilities 0.25 and 0.85, and service cost 0.25. The finite-state oracle and switch-time checks validate the disclosed transition system; they do not establish broad external validity or a general algorithm ranking. The diagnostic materials are provided in Online Resource 3.

S5. Resolution-cost extensions

The extensions use frozen v3-selected configurations and are reported as observed costs for disclosed task--algorithm cells. They are not pooled into the v3 family and do not estimate prospective power.

Table S8. Resolution-cost extension summary.

Extension	New or reused sequence	Rows	Decision summary	Evidential scope
Queue follow-up	Reuses the v3 20-seed prefix, then adds 380 seeds	800 total queue rows	Inconclusive at 200; DQN--A2C resolves at 400	One post-v3 controlled-task comparison
v4	New seeds 30000--30399	6,000	One queue DQN--A2C superiority decision at 200; 14/15 unresolved at 400	Same-family new-seed budget observation
v5	New seeds 30400--31399	15,000	Four terminal decisions by 1,000; 11/15 unresolved at 1,000	Same-machine resolution-cost diagnostic
v6 task-family extension	New tasks CliffWalking-v1 and fixed FrozenLake-v1, seeds 41000--41199	1,440	Six comparisons unresolved at 200	Same-machine cross-family execution and scope diagnostic

The v4 and v5 audits verify seed non-overlap, frozen configurations, bounded metrics, row counts, stop traces, and recomputed decisions. Because the v5 runner retained the complete sequence, a prefix decision is a sample-resolution location, not a measured wall-clock saving. These extensions do not support a universal seed requirement, a general PPO/DQN/A2C ranking, or external reproducibility.

The v6 extension uses ten validation seeds per candidate and 200 test seeds per cell for two additional fixed discrete tasks. CliffWalking-v1 evaluation is explicitly capped at 100 steps because its Gymnasium spec has no TimeLimit wrapper; FrozenLake-v1 is fixed to the 4 x 4 slippery map. The raw artifact contains 240 validation rows and 1,200 test rows. An independent audit recomputes all 24 finite-look decisions from the raw CSV, and all six remain max_budget_inconclusive at 200 seeds. The extension broadens execution scope only; it is not pooled with v3 and does not support cross-family power, algorithm ranking, external reproducibility, or scientific replication. Its preregistration, report, raw rows, and audit are provided in Online Resource 3.

S6. Reproducibility materials

The package distinguishes three levels:

1. Same-machine repeatability: the independent replay script reproduces all

15 v3 selected-test summaries with recorded mean absolute normalized-metric differences of 0.0.

2. Different-machine reproducibility: demonstrated by the returned Windows

AMD64 sealed rerun. All 15 selected configurations and terminal decisions match; 14/15 cells have zero seed-level differences and the remaining Acrobot-v1:DQN cell has mean absolute difference 0.00105 and maximum difference 0.01640. The raw output and full audit are provided in Online Resource 6.

3. Independent-team identity: not verified by the returned handoff. The

rerun therefore does not establish scientific replication on a new task family.

4. Scientific conclusion replication: not demonstrated; it would require a

new sealed study or independently selected task family.

The source archive contains raw seed-level CSV files, JSON summaries, source hashes, package versions, preregistration files, audit scripts, and exact commands. The data and code supporting this paper are supplied as Online Resource 2 and Online Resource 3 and Online Resource 4 and Online Resource 5 and Online Resource 6. The fixed release is v3.0.5; the archive series is available at https://doi.org/10.5281/zenodo.23949004.

S7. Audit materials

Executable audit code, environment specifications, expected outputs, and reproduction instructions are included with the code and provenance materials in Online Resource 5. These checks audit the stored package; they do not perform an external replication or establish journal acceptance.

S8. Data dictionary

Each seed-level row stores the task ID, algorithm ID, split, configuration ID, training seed, checkpoint returns, normalized aggregate metric, raw aggregate metric, software/runtime metadata, and provenance fields needed to reconstruct the selected procedure. Validation rows identify all candidates; test rows identify the frozen selected configuration. Summary JSON files store means, interval endpoints, look traces, decisions, and terminal states. The raw CSV is authoritative for row-level values; generated tables are derived views. For a task--algorithm cell used in more than one unordered pair, a training row is generated once and the stored row is reused by each applicable pair. Pair decisions can therefore lock at different looks while the shared cell continues to receive rows for any other open pair. This is why the reported training-run counts are cell counts rather than pair counts.

S9. Supplementary limitations

The calibration grid cannot replace a proof. The sensitivity analyses are conditional on stored rows and fixed configurations. The continuous, partial-observation, and v6 task-family runs are scope diagnostics rather than domain-wide evidence. The queue is self-built, and the v6 CliffWalking results are at the normalized floor under its short training budget. The larger-budget extensions are motivated by the v3 boundary and use the same machine. No supplementary item closes the four submission gates for external reruns, public archiving, independent literature screening, or author declarations.

S10. Supplementary claim--evidence map

Table S9. Supplementary claim--evidence map.

Supplementary claim	Evidence	Permitted wording
The conservative boundary persists on the disclosed calibration grid	80 cells, 240,000 replicates, stored JSON and report	"In the tested grid, no terminal decision occurred by 20 observations."
The v3 conclusion is insensitive to the tested normalization schemes	11 stored-bound schemes and recomputed decisions	"The stored comparisons remained unresolved under these post-hoc schemes."
Resolution cost varies within the frozen task family	Queue, v4, and v5 audits	"Observed resolution cost varied across disclosed frozen cells."
The workflow executes on two additional fixed discrete tasks	V6 task-family audit, raw 1,440-row data set, and preregistration (Online Resource 3)	"The extension broadens execution scope; all six comparisons remained unresolved at 200 seeds."
Seed dependence can invalidate nominal coverage	Six block-correlated bounded surrogate conditions	"The stress test illustrates the conditional independence requirement; it does not estimate dependence in RL."
Same-machine replay is exact at recorded precision	15 replay differences equal 0.0	"The package supports same-machine cross-script repeatability."
Different-machine reproducibility is established for the frozen protocol	Windows AMD64 sealed rerun, raw output, source/output hashes, and 15/15 matched decisions	Claim only the disclosed protocol/task family; do not call it scientific replication or independent-team evidence.
Twenty-validation-seed tuning is universally sufficient	The prospective extension changes 7/15 selections relative to v3	Do not claim sufficiency; report observed selection sensitivity for the fixed grid and task family.
A universal seed budget or algorithm ranking is established	No population or cross-machine evidence	Do not claim it.

S11. V7 new-seed method comparison

V7 was frozen after the V1--V6 results had been inspected and therefore is a post-hoc prospective validation rather than a confirmatory analysis of the original preregistration. It evaluates the same five-task panel, three algorithms, frozen configurations, and 15 comparisons on new training seeds 31,400--32,399 (15,000 successful training records). The original Hoeffding interval and eight variance-adaptive or sensitivity methods were evaluated at 200, 400, 600, 800, and 1,000 seeds under the same comparison-level alpha.

At 1,000 seeds, the main Gamma-mixture empirical-Bernstein confidence sequence was narrower than the original Hoeffding interval in all 15 comparisons. Its mean and median paired width differences were -0.185260 and -0.179334. The main Gamma setting produced four A-superior, two B-superior, five precision-only, and four unresolved comparisons; Hoeffding produced three A-superior, one B-superior, and eleven unresolved comparisons. The shared-arm counterfactual prefixes were 11,800 and 14,600 training runs, respectively, but all 15,000 runs were completed, so these are not realized compute savings.

The rho x 0.1 sensitivity produced six superiority and five precision-only decisions. The rho x 10 sensitivity produced six total terminal decisions, no precision-only decisions, and nine unresolved comparisons. A 50-seed block sensitivity removed all four original Hoeffding superiority decisions, but the blocks were not shown to be independent and therefore do not repair the primary assumption. The registered-parameter S23.3 result is an analytic-boundary check, not evidence that dependence is harmless. Complete method counts, widths, prefix calculations, V5/V7 comparisons, and raw files are listed in the S12 manifest.

This section supports only a fixed-panel interval-width and tuning-sensitivity description. It does not establish unknown-distribution coverage, task-population generalization, a universal seed budget, algorithm rankings, independent-team replication, or realized compute savings.

S12. Online Resource manifest

The following materials correspond to release v3.0.5. The Zenodo DOI https://doi.org/10.5281/zenodo.23949004 identifies the archive series; the release fixes the version used here.

Table S10. Online Resource manifest.

Resource	Content	Download
Supplementary narrative and tables	Extended methods, diagnostics, limitations, and claim--evidence map	Online Resource 1
Primary v3 data package	Preregistration, row-level metrics, configurations, and summaries	Online Resource 2
Supplementary experiment package	V4--V7 extensions, sensitivity analyses, and scope diagnostics	Online Resource 3
Dependence diagnostics	Controlled seed-dependence stress test and known-block fallback	Online Resource 4
Code and provenance package	Runner, audit scripts, environment metadata, references, and checksums	Online Resource 5
External rerun package	Windows raw data, audit, summaries, runtime metadata, logs, and checksums	Online Resource 6